

A triangle contradicts the proposed flag-complex chromatic conversion

Disproof of Conjecture 00000004227

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Abstract

The clique complex of the complete graph K_3 is a filled triangle. It has exactly one spanning two-tree, also of torsion weight one. Its clique number is 3, while its chromatic polynomial $P(q) = q(q-1)(q-2)$ satisfies $P(-1) = -6$. Thus the stated conversion with constant one fails. Absolute values or the usual parity correction give 6, and also fail. In fact no evaluation at a negative integer, even after taking its absolute value, equals the two-tree count in this example.

The statement and its evaluation point

The source proposes a conversion of the two-tree count of a graph's flag complex into an evaluation of its chromatic polynomial at a negative integer, with conversion constant 1 and point determined by the clique number minus 2. Write $\omega(G)$ for the usual clique number, the maximum size of a clique. The corresponding negative point is $-(\omega(G) - 2)$. We test that claimed unit conversion directly. The paper also checks absolute values and the opposite sign of the point, so the contradiction does not depend on a sign convention. No unspecified linear combination of several evaluations is assumed to be the proposed formula; the source gives no such combination or its coefficients.

A flag complex of dimension exactly two

Let $G = K_3$ on vertices $0, 1, 2$. Every subset of the vertex set is a clique, so its clique complex K consists of all subsets of $\{0, 1, 2\}$. It contains exactly three edges and the one triangular face 012 . In particular it is a flag complex of dimension 2, and $\omega(G) = 3$. There are no higher-dimensional faces to truncate.

We use simplicial spanning two-trees: a subcomplex keeps the entire one-skeleton, has no rational two-cycle, and every one-cycle is a boundary of a selected triangular two-chain. These are the chain versions of the standard homological spanning-tree conditions; see Duval–Klivans–Martin, Section 1.2. The direct calculation below also settles the integral homology and therefore the usual torsion weighting.

Orient the edges as $01, 02, 12$ and the face as 012 . The boundary matrices are

$$B_1 = \begin{pmatrix} -1 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & 1 \end{pmatrix}, \quad B_2 = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}.$$

They come from $\partial_1[u, v] = [v] - [u]$ and $\partial_2[a, b, c] = [b, c] - [a, c] + [a, b]$, so $B_1 B_2 = 0$. For $z = (z_0, z_1, z_2)^T$, the equation $B_1 z = 0$ says

$$z_1 = -z_0, \quad z_2 = z_0, \quad z = z_0(1, -1, 1)^T.$$

Consequently $\ker B_1 = \text{im } B_2$, and B_2 is injective.

Theorem 1. *The number of spanning two-trees of K is $\tau_2(K) = 1$. The torsion-weighted count is also 1.*

Proof. Keeping its unique triangular face gives an injective top boundary and fills every one-cycle by the displayed formula. It is therefore a spanning two-tree. Omitting that face leaves the full triangular one-skeleton with the nonzero one-cycle $(1, -1, 1)^T$ and no two-face to fill it, and hence is not a spanning two-tree. These are the only two possible face selections.

The same calculation holds with integral coefficients: every integral cycle is the boundary of the integer two-chain with coefficient z_0 . Thus the unique tree has $H_1(K; \mathbb{Z}) = 0$, and its weight $|\text{Tor } H_1(K; \mathbb{Z})|^2$ equals 1. The count is unchanged if that standard weighting is intended. $\square$

The actual chromatic polynomial

A proper coloring of K_3 assigns distinct colors to all three vertices. For $n \geq 3$ available labeled colors there are $n(n-1)(n-2)$ such colorings; for $n = 0, 1, 2$ there are none and this polynomial also vanishes. Thus, for every nonnegative integer n , the polynomial

$$P(q) = q(q-1)(q-2)$$

evaluates to the actual number of proper colorings. A polynomial over $\mathbb{Q}$ is determined by its values at all nonnegative integers, so this identifies the chromatic polynomial, including its evaluations at negative arguments.

Since $-(\omega(G) - 2) = -1$, the claimed conversion gives

$$\boxed{\tau_2(K) = 1 \neq -6 = P(-1)}.$$

Taking $|P(-1)|$, or the parity-corrected value $(-1)^3 P(-1)$, gives $6 \neq 1$. If the phrase “clique number minus two” is read as the positive argument instead, $P(1) = 0 \neq 1$. More generally, for every integer $m \geq 1$,

$$P(-m) = -m(m+1)(m+2) \leq -6, \quad |P(-m)| \geq 6.$$

No negative integer evaluation, with either overall sign, equals 1. This rules out changing the negative evaluation point as a repair of the unit-conversion claim on this graph. Different normalization factors or unspecified combinations would constitute different formulas.

Formal correspondence and reproduction

The supplied Lean project uses Mathlib’s actual `SimpleGraph`, clique predicate, clique number and proper-coloring type. The complex is defined by filtering vertex subsets by the graph’s clique predicate. Its downward closure, exact dimension and the unique two-face enumeration are checked.

An explicit equivalence identifies proper colorings with embeddings of the three-vertex type into the color type. The theorem `chromatic_counts` therefore establishes the counting property for *every* natural color count; `chromatic_unique` proves uniqueness using infinitely many evaluation points. The polynomial is not substituted for a graph object without proving its defining property. This definition-level characterization is used because the pinned Mathlib version does not provide a general graph chromatic-polynomial object.

The actual oriented simplices define both boundary linear maps over $\mathbb{Q}$. The predicate `IsSpanningTwoTree` uses their kernels, fillings and the support of a selected-face chain. The theorem `tree_iff_full` classifies every mask, and the resulting cardinality is proved to be 1. The final theorems compare that count with the polynomial evaluation, its absolute value and

the alternate positive argument. `no_negative_integer_unit_conversion` treats all negative integers. The integral-homology interpretation is proved above by the same explicit integer filling; no imported Lean homology theorem is claimed.

The project pins Lean 4.19.0 and Mathlib revision `c44e0c8ee63ca166450922a373c7409c5d26b00b`. In `lean/`, run

```
lake exe cache get
lake build
lake env lean -DwarningAsError=true Main.lean
```

The `cache` command fetches official dependencies when absent. The submission itself is rebuilt in a fresh directory, and its printed theorem dependencies use only Lean’s standard foundational axioms. No proof placeholders, additional axioms or native numerical oracles are used. Run `tectonic main.tex` to regenerate the PDF. The exact bilingual source, its provenance, and build and review records are included. This work received a separate adversarial self-review pass by its author; no independent reviewer is claimed.

References

1. The Last Math Competition, Conjecture 00000004227, exact source reproduced in this submission.
2. A. M. Duval, C. J. Klivans and J. L. Martin, *Simplicial matrix-tree theorems*, Section 1.2, for the simplicial tree convention.